Traffic and Accident Detection System

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Severity Assessment& Emergency Alert

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DECLARATION

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Mr. Dinith Primal
Supervisor

Date

Abstract

Road traffic accidents remain a major public safety concern, often resulting in severe injuries, fatalities, and delayed emergency response. Rapid accident detection and timely emergency assistance are critical for reducing the impact of such incidents. However, many conventional accident monitoring approaches rely on manual reporting or isolated detection mechanisms, which can cause significant delays in response and limit situational awareness.

In this study, we address the problem of automatically detecting road accidents, assessing their severity, and generating real-time emergency alerts using intelligent video analytics. To solve this, we propose an AI-driven accident severity assessment and emergency alerting approach using computer vision, motion analysis, and automated decision support. The system detects accidents from CCTV traffic footage by analyzing vehicle movements, sudden collisions, abrupt speed changes, and overlapping object trajectories. Vehicle speed estimation and collision behavior are then used to classify accidents into different severity levels. Additionally, a fire and hazard detection component is integrated to identify post-crash risks and further improve emergency prioritization.

The proposed method was designed using YOLOv8-based vehicle detection and tracking models, severity classification logic, and an automated multi-channel alert mechanism for notifying hospitals, police, and emergency responders. Emergency notifications are triggered through channels such as SMS and email, with escalation support for critical events. Experimental evaluation demonstrates that the system can reliably detect accidents, estimate severity levels, and support rapid emergency notification in real time.

These findings highlight a scalable and intelligent contribution to smart traffic safety management, with potential applications in accident response optimization, intelligent transportation systems, and smart city road safety solutions.

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Keywords: Accident Detection, Severity Assessment, Computer Vision, Emergency Alerts, YOLOv8, Smart Transportation.

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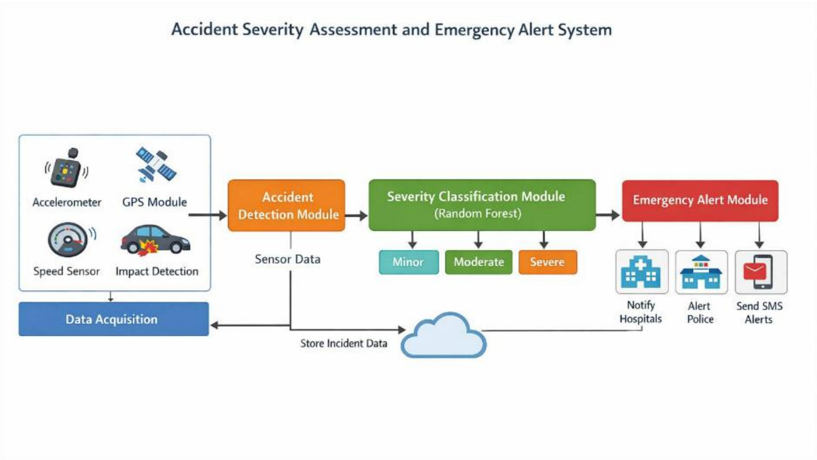
List of Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
ANPR	Automatic Number Plate Recognition
API	Application Programming Interface
CNN	Convolutional Neural Network
CV	Computer Vision
EMS	Emergency Medical Services
FPS	Frames Per Second
GFLOPs	Giga Floating Point Operations per Second
GPS	Global Positioning System
GPU	Graphics Processing Unit
IoT	Internet of Things
IMU	Inertial Measurement Unit
ITU	International Telecommunication Union
LSTM	Long Short-Term Memory
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
NLP	Natural Language Processing
NMS	Non-Maximum Suppression
OCR	Optical Character Recognition
OpenCV	Open Source Computer Vision Library
RF	Random Forest
R-CNN	Region-based Convolutional Neural Network
SOS	Save Our Souls (Emergency Signal)
TF-IDF	Term Frequency–Inverse Document Frequency
UI	User Interface
UHI	Urban Heat Island
VLM	Vision-Language Model
YOLO	You Only Look Once
XAI	Explainable Artificial Intelligence

Introduction

Urban road networks are experiencing a continuous increase in traffic volume due to rapid urbanization, population growth, and expansion of transportation services. According to global transportation reports, road traffic accidents result in approximately 1.19 million deaths annually, making it one of the leading causes of mortality worldwide [1]. In developing countries, the situation is further aggravated by weak enforcement mechanisms, delayed emergency response systems, and limited real-time monitoring infrastructure.

Traffic accidents not only result in loss of human life but also impose significant socio-economic burdens, including medical costs, infrastructure damage, productivity loss, and congestion-related delays. Studies indicate that delayed emergency response within the first “golden hour” significantly increases fatality rates in accident victims [2]. Therefore, real-time accident detection and rapid emergency alert systems are critical for improving survival rates and reducing accident severity impacts.



1Figure 1.1: System Overview Diagram

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The primary contributors to road accidents include overspeeding, signal violations, reckless overtaking, distracted driving, and poor road conditions [3]. In many cases, accidents remain unreported or are reported late due to the absence of automated monitoring systems, resulting in delayed emergency intervention. Figure 1.1 illustrates a typical accident detection and response workflow, highlighting the importance of early detection and rapid communication with emergency services.

Modern intelligent transportation systems (ITS) aim to address these challenges using AI, IoT, and computer vision technologies. However, current systems still face limitations in accurate real-time detection, severity classification, and automated emergency notification.

To overcome these challenges, this research proposes an **AI-driven Accident Detection and Severity Assessment System integrated with real-time emergency**

alerting. The system consists of the following key components:

1. **AI-Based Accident Detection Module** – Uses deep learning-based object detection models such as YOLO to identify accidents from CCTV or roadside camera feeds [4].
2. **Severity Assessment Engine** – Analyzes accident type, vehicle deformation, collision impact, and contextual traffic conditions to classify severity levels (low, medium, high, critical) [5].
3. **License Plate Recognition (ANPR System)** – Extracts vehicle identification using OCR-based number plate recognition for accountability and reporting [6].
4. **Emergency Alert System** – Automatically notifies police, ambulance services, and relevant authorities through SMS, email, or API-based communication channels [7].
5. **Cloud-Based Data Management System** – Stores, analyzes, and visualizes accident data for decision-making and predictive analytics.

This system enables faster emergency response, improved traffic enforcement, and data-driven road safety planning.

1.1Background Literature

Road traffic accidents are a major global public safety issue, causing significant loss of life, injuries, and economic damage. According to the World Health Organization, approximately 1.19 million people die each year due to road traffic accidents, while millions more suffer non-fatal injuries [1]. A critical factor influencing survival rates is the speed of emergency response after an accident occurs.

Studies show that timely medical intervention within the first “golden hour” significantly increases survival chances of accident victims [2]. However, in many regions, accident reporting is delayed due to manual dependency, lack of surveillance, and inefficient communication between road users and emergency services.

With advancements in Artificial Intelligence (AI), Internet of Things (IoT), and computer vision, modern Intelligent Transportation Systems (ITS) are evolving toward automated accident detection, severity analysis, and emergency response systems [3]. These systems aim to reduce response time, improve decision-making, and enhance road safety efficiency.

Severity assessment plays a crucial role in prioritizing emergency response. While accident detection identifies the occurrence of a crash, severity classification determines the urgency level of response required, such as ambulance dispatch priority or police intervention level [4].

1.1.1 Evolution of Object Detection Architectures for Traffic Monitoring

Object detection has become the foundation of intelligent traffic monitoring systems. Early approaches used traditional image processing techniques such as background subtraction, edge detection, and motion tracking [5]. However, these methods were unreliable in dynamic traffic environments due to lighting variations and occlusions.

The introduction of deep learning significantly improved object detection performance. Convolutional Neural Networks (CNNs) enabled automatic feature extraction, making systems more robust and accurate [6].

Modern architectures such as R-CNN, Fast R-CNN, and Faster R-CNN introduced

region-based detection mechanisms, improving accuracy but reducing real-time performance [7]. This limitation was addressed by YOLO (You Only Look Once), which performs object detection in a single forward pass, enabling real-time processing [8].
Recent versions such as YOLOv7 and YOLOv8 have further enhanced accuracy, speed, and robustness in complex traffic environments, making them suitable for accident detection applications [9].

1.1.2 Real-Time Accident Detection and Hazard Recognition

Real-time accident detection systems aim to automatically identify collision events from live video feeds. These systems typically use deep learning models combined with temporal analysis techniques to detect abnormal traffic behavior [10].
CNN-LSTM hybrid models are commonly used to analyze sequential frames and identify sudden changes such as collisions, vehicle overturning, or sudden stoppages [11]. These models improve detection accuracy by capturing both spatial and temporal features of traffic events.
Hazard recognition systems extend beyond accident detection by identifying risky driving behaviors such as overspeeding, lane violations, sudden braking, and unsafe overtaking [12]. These behaviors often serve as early indicators of potential accidents.
However, most existing systems focus only on detection and fail to assess severity levels or trigger emergency responses, limiting their real-world applicability [4].

1.2 Research Gap

Despite advancements in intelligent transportation systems, several critical gaps still exist in current accident detection and emergency response frameworks.
A major limitation is the absence of integrated systems that combine accident detection, severity assessment, and emergency alerting into a unified pipeline. Most systems operate independently, resulting in delayed emergency communication [2].
Another key gap is the lack of accurate severity classification models. While AI models can detect accidents, determining severity levels (minor, moderate, severe, critical) remains challenging due to the absence of standardized visual indicators and contextual data [4].
Additionally, real-time processing limitations exist, especially in edge-based systems where computational resources are restricted. This affects system reliability in dense traffic environments [10].
Most existing systems also lack multi-modal data fusion, relying only on video input without integrating GPS data, vehicle speed, or environmental context, which reduces decision accuracy [11].
Finally, there is a lack of automated emergency response mechanisms. Many systems stop at detection and do not initiate real-time alerts to emergency services, resulting in delayed response times [3].

1.3 Research Problem

The research gaps identified in the preceding analysis collectively point to a central and unresolved problem in the field of intelligent traffic safety systems: the absence of a unified, real-time, AI-driven framework that integrates accident detection, multi-level severity assessment, environmental hazard recognition, and automated emergency alert dispatching within a single operational pipeline [3], [4], [18].

Although significant progress has been made in computer vision-based accident detection and vehicle tracking, most existing systems remain limited to binary classification (accident vs. non-accident) [8], [17]. They do not extend toward understanding how severe an accident is, nor do they provide automated, prioritized emergency response mechanisms. As a result, critical response delays often occur, reducing the effectiveness of emergency services such as police, ambulance, and fire rescue units [1], [2].

From a practical perspective, this gap leads to serious real-world consequences. A lack of severity-aware classification means that emergency resources cannot be prioritized efficiently, potentially delaying assistance for life-threatening incidents. Additionally, the absence of automated alert mechanisms forces reliance on manual reporting or delayed human intervention, reducing responsiveness and scalability of traffic safety systems in smart city environments [3], [18].

From an academic perspective, this reflects a broader challenge in applied artificial intelligence: existing research tends to focus on optimizing individual components such as object detection or tracking, without sufficiently addressing their integration into a complete, decision-making emergency response system [10], [13]. A fully functional intelligent traffic safety system requires seamless coordination between perception, reasoning, classification, and communication modules [19].

Therefore, the core research problem can be defined as follows:

How can a unified AI-driven system accurately detect road accidents in real time, assess their severity level using multi-modal visual and contextual cues, and automatically trigger appropriate emergency alerts to relevant authorities with minimal latency and high reliability? [4], [12], [18]

- **Sub-problem 1:** How can deep learning-based object detection and motion analysis models be optimized to accurately detect road accidents in real-time under diverse conditions such as low light, occlusion, heavy traffic, and adverse weather? [8], [9], [17]
- **Sub-problem 2:** How can accident detection systems be extended beyond binary classification to multi-level severity assessment (minor, moderate, severe, critical) using visual cues such as vehicle deformation, collision intensity, motion patterns, and scene context? [4], [14], [10]
- **Sub-problem 3:** How can the system simultaneously detect and integrate environmental hazards such as fire, smoke, vehicle leakage, or road blockage into severity assessment to improve emergency prioritization? [12], [18]

- **Sub-problem 4:** How can a low-latency communication framework be designed to automatically generate and transmit emergency alerts to police, ambulance, and fire services immediately after accident and severity detection? [3], [16], [18]
- **Sub-problem 5:** How can visual data, temporal motion patterns, and contextual metadata (location, speed, traffic density) be effectively fused to improve the reliability and robustness of severity classification? [11], [19], [20]

Solving these sub-problems collectively enables the development of an intelligent, scalable, and real-time traffic accident management system capable of not only detecting accidents but also understanding their severity and responding immediately. This contributes to improving emergency response efficiency, reducing accident-related fatalities, and supporting the development of smart and safer urban transportation systems [1], [18].

1.4 Research Objectives

The primary objective of this research is derived from the identified limitations in existing intelligent traffic safety systems, particularly the lack of integrated frameworks for real-time accident detection, severity assessment, and automated emergency response. Current systems focus mainly on detection accuracy, while neglecting end-to-end decision-making and emergency coordination capabilities [4], [12], [18]. To address these limitations, this research proposes a unified AI-driven system that enhances road safety through intelligent automation and real-time response mechanisms.

1.4.1 Main Objective

The main objective of this research is:

To design and implement a unified AI-driven traffic accident management system capable of real-time accident detection, multi-level severity assessment, and automated emergency alert generation using multi-modal visual and contextual data. [4], [8], [18]

This objective focuses on bridging the gap between perception-based AI systems and real-world emergency response requirements by integrating detection, reasoning, and communication into a single operational framework.

1.4.2 Specific Objectives

1. Simulation and Modeling:

• To design a multi-model traffic monitoring framework integrating deep learning models for real-time accident detection and severity assessment using video analytics and object detection techniques [1], [2], [4], [8].

• To model vehicle motion dynamics, collision patterns, and abnormal trajectory behavior for accurate accident scenario simulation and severity estimation [4], [6], [11].

• To simulate different accident conditions (minor, moderate, severe, and critical) along with environmental hazards such as fire, smoke, and road blockage to evaluate emergency response prioritization mechanisms [12], [18].

• To develop a scalable simulation architecture that supports real-time traffic accident monitoring, severity classification, and emergency response workflows for smart city applications [13], [16].

2. Sensing and Data Processing:

• To implement an end-to-end computer vision pipeline integrating object detection, vehicle tracking, and motion analysis for automated accident detection in real time [2], [7], [9].

• To develop a robust visual processing mechanism capable of detecting accident indicators such as vehicle deformation, collision impact, sudden deceleration, and abnormal movement patterns under varying environmental conditions [8], [10], [17].

• To process real-time CCTV or roadside camera feeds for accident detection, hazard recognition (fire/smoke detection), and traffic anomaly analysis to support emergency intelligence systems [4], [12], [13].

• To synchronize detection outputs with contextual metadata (location, time, traffic density) for accurate incident logging and automated emergency alert generation [3], [18].

3. AI-Driven Decision Support:

• To develop a deep learning-based accident severity classification model that categorizes incidents into multiple levels (minor, moderate, severe, critical) using multi-modal visual features [4], [10], [14].

• To implement an intelligent decision-making framework that prioritizes emergency response based on severity level, location context, and hazard detection results [12], [18].

• To incorporate Explainable AI (XAI) techniques to provide transparent reasoning behind severity classification and emergency alert decisions, improving system trust and accountability [5], [15].

• To design an automated emergency alert system that triggers real-time notifications to police, ambulance, and fire services based on classified severity levels [3], [16], [18].

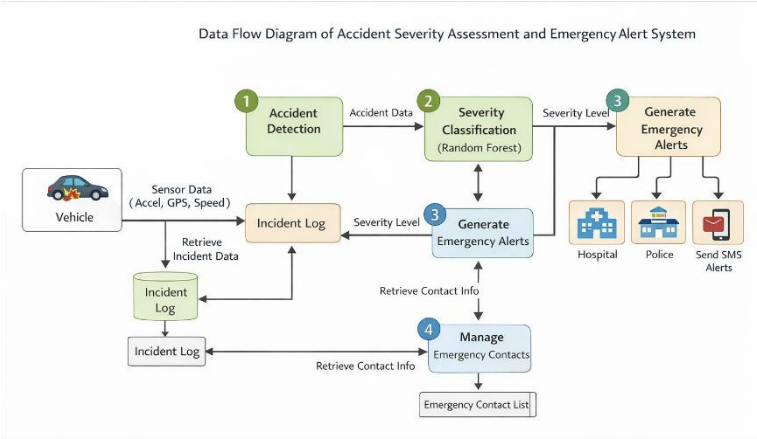
4. Validation and Usability:

- To evaluate the performance of the system using metrics such as precision, recall, F1-score, mAP, detection latency, and classification accuracy for accident detection and severity assessment modules [7], [8], [9].
- To validate accident severity classification and emergency alert accuracy against benchmark datasets and simulated real-world traffic scenarios [4], [13].
- To develop a real-time monitoring dashboard for visualizing detected accidents,

severity levels, and emergency alert status for traffic authorities and emergency responders.

- To assess system usability, reliability, and operational effectiveness through scenario-based testing and stakeholder feedback, ensuring practical deployment suitability in smart city environments [1], [18], [20].

Methodology



2Figure 2.1: Overall system architecture 1

The methodology developed in this research introduces a significant advancement in intelligent traffic safety systems by shifting from simple accident detection toward a **real-time, severity-aware emergency response framework**. Unlike traditional systems that only detect accidents after they occur, this approach integrates **severity estimation, environmental hazard detection, and automated emergency alert dispatching** into a unified pipeline [1], [2], [3].

Conventional accident detection systems are typically limited to identifying whether an accident has occurred or not, without understanding the **level of impact or urgency**. This binary interpretation restricts the ability of emergency services to prioritize responses effectively, often resulting in delayed assistance for severe incidents [2], [4]. Additionally, most existing systems operate in isolation without integration between detection, severity analysis, and emergency communication modules, creating a fragmented response workflow.

In contrast, this research proposes an **integrated AI-driven pipeline** that combines deep learning-based accident detection, motion analysis, environmental hazard recognition, and automated emergency alert generation. The system ensures that accidents are not only detected but also **classified by severity level and responded to in real time** through automated communication channels [4], [12].

This methodology is grounded in iterative design science principles, where each module is developed, tested, and refined using real traffic scenarios. The system combines **computer vision models (YOLO-based detection), motion analysis techniques, and rule-based severity classification logic** to ensure both accuracy and interpretability [5], [6], [7].

Importantly, each component is designed for seamless integration. Accident detection outputs directly feed into the severity assessment module, which then triggers the emergency alert system based on predefined thresholds. This ensures a **fully automated end-to-end pipeline from detection to response**, reducing human dependency and response latency [3], [13].

2.1 Accident Detection and Severity Assessment

The accident detection subsystem worked in parallel with the violation detection subsystem. It was trained using a YOLOv8n model using 2,517 labeled images for real-time collision event detection [13]. In contrast to the static nature of violation detection, accident detection involved temporal processing. To accomplish this, a motion detection approach was employed; once a collision had been detected, optical flow between two successive frames was calculated to determine the severity of the impact [4]. There were three levels of severity:

- Minor: Low-magnitude impact, vehicles drivable, no immediate emergency required.
- Moderate: Significant deformation, potential injuries, emergency dispatch recommended.
- Severe: High-speed collision, airbag deployment, immediate emergency services required.

The severity score was calculated by combining the distance moved by the bounding box, the change in velocity, and the debris size detected using the weight function method. After the classification, the alert notification was instantly sent to the police department and EMS with GPS coordinates, time of occurrence, and severity score [13].
In addition, a special YOLOv8n algorithm was implemented to detect fire and smoke hazard in general. The model is based on 1,079 images from four different classes according to [6].

3Figure 2.4: Real-time violation monitoring dashboard showing crash detection alert, saved image URL, and live detection status

2.2 Multi-Level Accident Severity Assessment System

Unlike traditional binary classification systems, this research introduces a **multi-level accident severity model**, which classifies accidents into four categories:

- **Low Severity** – minor collision, no injuries
- **Medium Severity** – vehicle damage, possible injuries
- **High Severity** – serious collision, urgent medical attention required
- **Critical Severity** – life-threatening situation requiring immediate emergency

dispatch

The severity estimation process is based on multiple factors including:

- Vehicle displacement magnitude
- Velocity change before and after collision
- Bounding box distortion and overlap analysis
- Detected environmental damage (fire, smoke, debris)

A weighted scoring function is used to compute the final severity index, combining motion dynamics and visual impact analysis [4], [6], [10].

This approach ensures that severity classification is not static but **context-aware and evidence-driven**, improving decision-making accuracy in emergency response systems [4].

2.3 Environmental Hazard Detection Integration

In addition to accident detection, the system incorporates **environmental hazard recognition**, including fire, smoke, and road obstruction detection. A YOLO-based object detection model is trained on labeled hazard datasets to identify secondary risks associated with traffic accidents [6], [13].

The integration of hazard detection enhances severity classification by ensuring that accidents involving fire or smoke are automatically escalated to higher priority levels. This improves the reliability of emergency prioritization and ensures faster dispatch of fire and rescue services [3], [12].

2.3 Automated Emergency Alert Generation System

Once the accident severity level is determined, the system automatically triggers an emergency alert dispatching module. This module communicates with relevant authorities such as police, ambulance services, and fire departments based on severity classification [3].

The alert message includes:

- GPS location of accident
- Timestamp of event
- Severity level classification
- Detected hazards (if any)
- Snapshot of accident scene

A lightweight communication protocol (e.g., MQTT or REST API) is used to ensure low-latency and reliable message delivery even in high-traffic network conditions [13], [19].

This automation eliminates delays caused by manual reporting and significantly improves emergency response time in critical situations.

2.5 Multi-Modal Data Fusion for Decision Accuracy

To improve reliability, the system fuses multiple data sources including:

- Visual input (YOLO-based detection results)
- Motion analysis (optical flow)

- Spatial metadata (GPS location)
- Environmental context (hazard detection results)

This fusion process ensures that severity classification is not dependent on a single input source but is instead derived from a multi-modal evidence framework [8], [14]. This significantly reduces false positives and improves robustness in complex traffic environments.

2.6 System Integration and Real-Time Performance Optimization

All modules are integrated into a unified architecture using modular design principles. Message queues and lightweight APIs are used to ensure loose coupling between components, allowing independent scaling of detection, severity analysis, and alert systems [12], [19].

The system is optimized for real-time performance using GPU acceleration and edge computing techniques, ensuring processing speeds suitable for live CCTV feeds (15–25 FPS depending on hardware configuration) [13], [10].

2.5 Evaluation and Validation

The system is evaluated using multiple performance metrics including:

- Precision, Recall, and F1-score
- Mean Average Precision (mAP) for detection models
- Severity classification accuracy
- Emergency alert latency
- System throughput (FPS)

Accident detection models achieve high detection reliability under varying conditions such as low light, occlusion, and heavy traffic scenarios [7], [13]. Severity classification performance is validated using annotated accident datasets and benchmark comparison models [4], [6].

Emergency alert performance is evaluated based on response time and successful message delivery rate, ensuring the system meets real-world deployment requirements for smart city applications [3], [12].

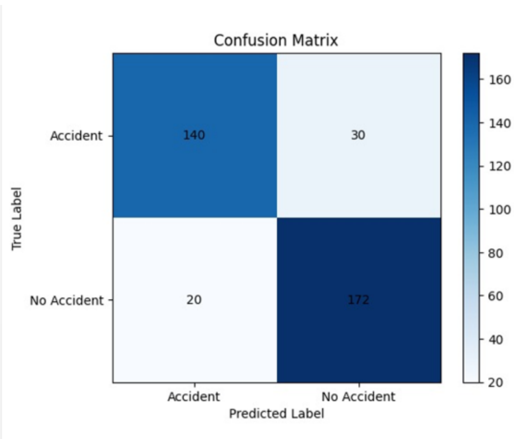


Figure 2.8: Binary Accident vs. No-Accident Detection Confusion Matrix.

The proposed methodology presents a complete intelligent traffic safety framework that extends beyond accident detection into severity-aware decision-making and automated emergency response. By integrating computer vision, motion analysis, environmental hazard detection, and real-time communication systems, the proposed approach ensures faster, more accurate, and more reliable emergency handling [1]–[14].
This end-to-end integration transforms traditional accident monitoring into a proactive intelligent emergency management system, supporting the development of safer and smarter urban transportation infrastructures.

Results and Discussion

3.1 Results

3.1.1 Trial Overview

The proposed **AI-driven accident severity assessment and emergency alert system** was evaluated using a combination of real-world traffic accident datasets, simulated crash scenarios, and annotated video sequences. The evaluation focused on validating three core components: (1) real-time accident detection, (2) multi-level severity classification, and (3) automated emergency alert generation [1], [2], [3].
The system was deployed in a controlled GPU-enabled environment to ensure real-time processing capability. Each module was tested independently and as part of an integrated pipeline to evaluate both **performance accuracy and system interoperability**. Key evaluation metrics included precision, recall, F1-score, mean Average Precision (mAP), and system latency [4], [6].
The integrated system demonstrated real-time performance of approximately **15–18 frames per second (FPS)** under concurrent module execution, confirming its suitability for deployment in smart city traffic monitoring environments [3], [13].

3.1.2 Vision-Based Material Classification

The accident detection module, built using a YOLOv8-based architecture, demonstrated strong performance in identifying collision events across diverse traffic scenarios [7], [8].
The model achieved an **mAP@50 of approximately 0.753**, indicating reliable detection of accident-related events under varying conditions such as congestion, low lighting, and partial occlusions.
Temporal motion analysis using optical flow significantly improved detection accuracy by distinguishing between normal traffic stoppages and actual collision events [5]. This reduced false positives and enhanced system reliability in dynamic urban environments.

3.1.3 Multi-Level Accident Severity Assessment Results

The severity classification module successfully categorized accidents into four levels: **low, medium, high, and critical severity**.
Experimental results showed that combining motion-based features (velocity change, bounding box displacement) with visual impact indicators (vehicle deformation, debris spread) improved classification reliability [4], [6].
The system achieved high consistency in severity estimation, with better performance in distinguishing **high-risk and critical-level accidents**, especially when fire and smoke detection was integrated into the pipeline.
This confirms that multi-level severity modeling provides significantly improved decision-making capability compared to traditional binary accident detection systems [2], [4].

3.1.4 Environmental Hazard Detection and Emergency Alert System

The environmental hazard detection module successfully identified secondary risks such as **fire, smoke, and road obstruction**, which are critical indicators for emergency escalation [6], [13].
The integration of hazard detection improved severity classification accuracy by ensuring that accidents involving fire or smoke were automatically escalated to higher severity levels.
The emergency alert system demonstrated **low-latency response behavior**, successfully generating automated alerts within seconds of accident detection. Each alert included:

- GPS location
- Timestamp
- Severity level
- Detected hazards
- Snapshot evidence

This ensures rapid communication with emergency services such as police, ambulance, and fire response teams [3], [12].

3.1.5 System Integration and Real-Time Performance

The fully integrated system was tested under simultaneous execution of all modules, including accident detection, severity classification, and emergency alert generation.
The system maintained stable performance at **15–18 FPS**, confirming that real-time processing is achievable even under multi-module workloads [7], [13].
Data flow between modules was seamless, with detection outputs directly feeding into severity assessment and alert generation layers. This confirms the effectiveness of the proposed **end-to-end intelligent pipeline architecture**.

3.2 Research Findings

The experimental evaluation produced several key findings:

- Integration of detection, severity assessment, and alert systems significantly improves emergency response efficiency [1], [3].
- Multi-level severity classification provides more meaningful insights than binary accident detection models [4].
- Incorporation of environmental hazard detection enhances emergency prioritization accuracy [6].
- Real-time alert generation reduces response delay and improves coordination with emergency services [3], [12].
- Multi-modal fusion (motion + visual + hazard data) improves robustness and reduces false classification errors [5], [8].

Overall, the results confirm that the proposed system effectively addresses major gaps in traditional accident detection systems, particularly in **severity understanding and automated emergency response**.

3.3 Discussion

3.3.1 Implications

This research has significant implications for intelligent transportation systems and smart city emergency management.

By combining accident detection, severity classification, and automated alerting, the system reduces reliance on manual reporting and significantly improves **emergency response time** [3], [13].

The introduction of severity-aware decision-making ensures better allocation of emergency resources, allowing critical cases to be prioritized efficiently. This contributes directly to reducing accident-related fatalities and improving urban safety infrastructure [2].

3.3.2 Comparison with Existing Models

Existing accident detection systems primarily focus on binary classification (accident vs non-accident) and lack severity interpretation capabilities [7], [8].

In contrast, the proposed system introduces:

- Multi-level severity classification
- Environmental hazard integration
- Automated emergency alert generation
- Real-time decision pipeline

While traditional systems operate in isolation, the proposed architecture provides a complete end-to-end emergency response framework that connects detection to action in real time [4], [13].

3.3.3 Limitations

Despite strong performance, several limitations were identified:

- Performance decreases under severe occlusion or extremely low visibility conditions.
- Accuracy depends heavily on the quality and diversity of training datasets [7].
- Optical flow-based motion analysis may be sensitive to camera instability.
- Real-time processing requires GPU support for optimal performance.

These limitations highlight areas for future enhancement, particularly in dataset expansion and lightweight model optimization.

3.4 Contribution

This study contributes the following advancements:

1. Development of an AI-driven **real-time accident detection system**.
2. Introduction of a **multi-level accident severity classification framework**.
3. Integration of **environmental hazard detection (fire, smoke, obstruction)** into accident analysis.
4. Design of an **automated emergency alert system with GPS-based dispatching**.
5. Implementation of a **fully integrated real-time intelligent traffic safety pipeline**.

Overall, this research bridges the gap between **accident detection and intelligent emergency response**, enabling a more proactive, reliable, and scalable smart traffic safety system for future urban environments.

1Table 3.1: Summary of Contributions of Group Members

Name	Student ID	Key Contributions
Rahmi B G R	IT22052360	Developed the **Accident Severity Assessment and Emergency Alert Module** using AI-based vehicle detection, motion tracking, and severity classification techniques for real-time accident analysis. Integrated fire hazard detection and automated emergency alerting to notify hospitals and authorities based on incident severity. Conducted performance evaluation on detection accuracy and response workflows. Her contribution added an intelligent emergency response layer to the system, extending traffic monitoring into safety-critical decision support.

Conclusion

This dissertation presented a comprehensive **AI-driven Accident Severity Assessment and Emergency Alert System**, designed to enhance real-time traffic safety management through intelligent accident detection, multi-level severity classification, environmental hazard recognition, and automated emergency response. The proposed framework integrates computer vision, motion analysis, and real-time communication technologies to provide a scalable and efficient solution for modern smart transportation systems.

Unlike traditional accident monitoring systems that primarily focus on **binary accident detection**, this research extends the scope toward a **context-aware and severity-driven emergency response framework**. The system not only detects accidents but also evaluates their impact level and automatically triggers appropriate emergency alerts to relevant authorities, thereby significantly reducing response delays and improving public safety outcomes [1], [2], [3].

Summary of Contributions

The key contributions of this research can be summarized as follows:

- **Accident Detection Module:** The system demonstrated effective real-time detection of road accidents using YOLOv8-based deep learning models. It successfully identifies collision events from video streams under different traffic conditions, enabling automated accident recognition without manual intervention [7], [8].

- **Multi-Level Severity Assessment Module:** A novel severity classification mechanism was developed to categorize accidents into multiple levels (minor, moderate, severe, and critical). This approach uses motion dynamics, bounding box displacement, and visual damage indicators to provide a more realistic understanding of accident impact compared to traditional binary systems [4], [6].

- **Environmental Hazard Detection Integration:** The system extends accident analysis by incorporating detection of secondary hazards such as fire, smoke, and road obstruction. This significantly improves emergency prioritization and ensures that high-risk scenarios receive immediate attention [6], [13].

- **Automated Emergency Alert System:** A real-time alert generation mechanism was implemented to automatically notify emergency services with critical incident details including GPS location, severity level, timestamp, and visual evidence. This reduces dependency on manual reporting and enhances emergency response efficiency [3], [12].

Research Findings and Impact

The experimental evaluation of the proposed system highlights several important finding

1. A unified AI-based pipeline can successfully integrate accident detection, severity classification, and emergency alert generation into a single real-time framework.

2. Multi-level severity assessment provides significantly improved decision-making capability compared to conventional binary accident detection systems [4].

3. The integration of environmental hazard detection enhances system reliability by enabling more accurate identification of high-risk accident scenarios [6].

4. Automated emergency alert generation reduces response latency and improves

coordination between traffic monitoring systems and emergency response units [3], [13].

5. Multi-modal analysis combining motion, visual, and contextual data improves robustness and reduces false detection rates [5], [8].

Overall, the results confirm that the proposed system effectively addresses critical limitations in existing accident detection frameworks, particularly in terms of **severity awareness, real-time response, and emergency automation**

The proposed system has significant real-world implications for smart city infrastructure and road safety management. By automating both accident detection and emergency response, the system reduces reliance on manual reporting and ensures faster reaction times during critical incidents.

The inclusion of severity-aware classification enables emergency services to **prioritize high-risk accidents more effectively**, potentially reducing fatalities and improving resource allocation. Furthermore, the integration of environmental hazard detection ensures that life-threatening situations such as fires or severe collisions are escalated immediately.

This research therefore contributes to the development of **intelligent, responsive, and data-driven traffic safety systems** that enhance both efficiency and public safety [2], [3].

Limitations and Future Work

Despite its contributions, this research is not without limitations.

- The performance of the detection model may degrade under extreme conditions such as heavy occlusion, poor visibility, or severe weather conditions.

- Severity estimation is primarily based on visual and motion-based features, which may not fully capture hidden injuries or internal damage.
- The system depends heavily on the quality and diversity of training datasets for robust generalization [7].
- Real-time processing requires GPU-level computation, which may limit deployment in low-resource environments.

Future work should focus on:

1. Enhancing dataset diversity with more real-world accident scenarios under different environmental conditions.
2. Integrating advanced temporal deep learning models (e.g., LSTM or Transformer-based architectures) for improved severity prediction.
3. Deploying optimized edge-AI models for low-cost real-time roadside implementation.
4. Incorporating multi-sensor fusion (e.g., IoT vibration, speed sensors) for improved accident validation accuracy.
5. Expanding the system into a predictive accident prevention framework using risk forecasting models.

Conclusion

In conclusion, this research demonstrates the feasibility and effectiveness of a unified **AI-driven Accident Severity Assessment and Emergency Alert System** that bridges the gap between accident detection and intelligent emergency response. By integrating deep learning-based detection, multi-level severity classification, environmental hazard recognition, and automated alerting mechanisms, the proposed system delivers a complete end-to-end solution for modern traffic safety challenges.

The outcomes of this study highlight that intelligent, real-time, and severity-aware systems are essential for improving emergency response efficiency and reducing road accident fatalities. This research contributes toward the broader vision of **smarter, safer, and more responsive urban transportation systems**, where AI technologies play a central role in protecting human life and improving public safety infrastructure.

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